

Trust the Card, Not the Resemblance

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KEY MESSAGE

A 5-point forfeit makes (Trust, Keep) the unique equilibrium when the card is misaligned.
In seven pairs Trust was higher on those cards, but Keep stayed near one half.

01 Question & model

Research question & verified gap

When the card is misaligned, what rule makes Trust an equilibrium even if SAME is public?
A screen may copy priority and tone. Both sides choose.
Gap: PS1 had no Promiser choice (Crawford and Sobel 1982).

Strategic model & foundations

Hidden actions: Keep or Break, and Trust or Verify.
Both lock first. SAME changes no payoff and cannot waive it.
Aligned 6/3; misaligned 6/10. Decider gets 6, -4, or 2.
Strict dominance; Nash (1950); Osborne and Rubinstein (1994).

Three complementary lenses

Game theory: (Verify, Break) flips to (Trust, Keep).
Social choice: Decider +4, Promiser -4; total stays 12.
Mechanism: forfeit 5 on a misaligned Break. SAME cannot waive it.

Auction or allocation comparison

Role: who is Promiser. Bids are private; the card is public.
The second bid is the forfeit, and it is paid only after Break.
Above 4: Trust, Keep. Below 4: Verify, Break. At 4: not unique.
Examples (9, 6) and (8, 3). Not played, and not the contribution.

02 Design & evidence

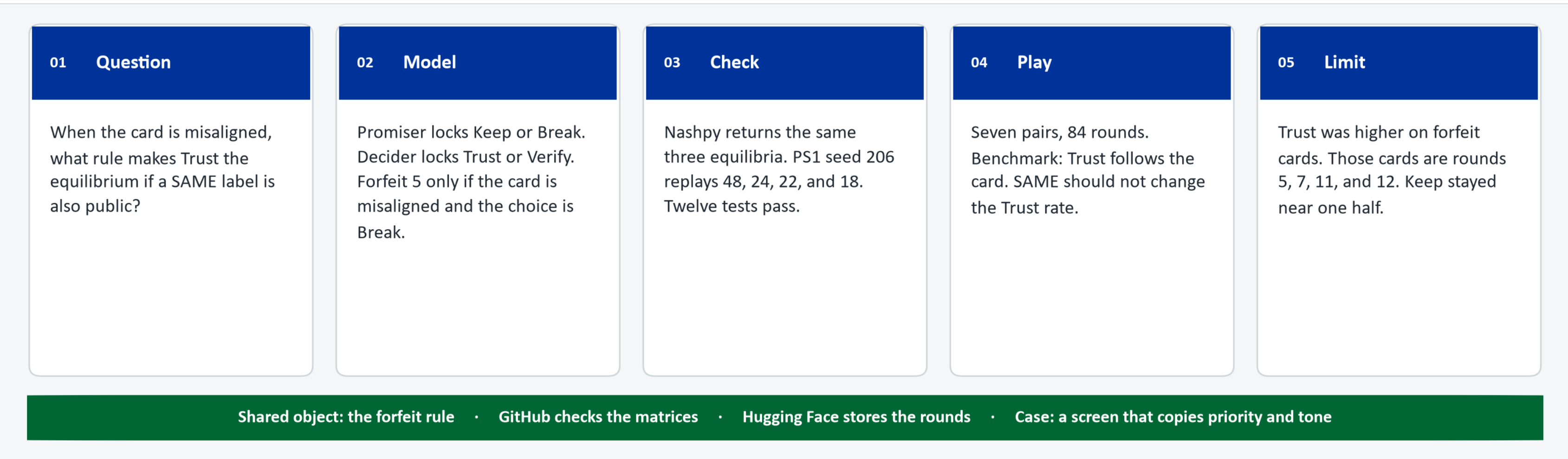


Figure 1. One chain: the shared question, the forfeit rule, the Nashpy check, seven pairs, and the order limit.

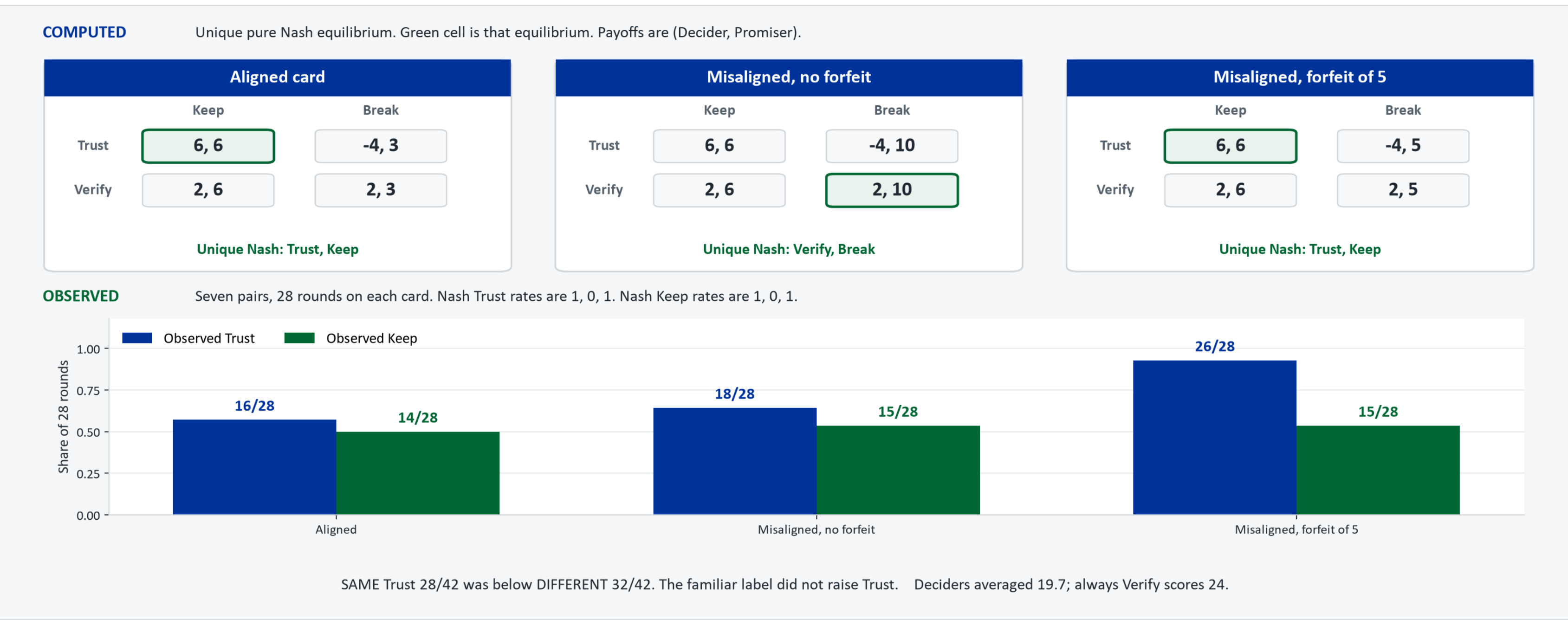


Figure 2. Computed Nash cells beside Trust and Keep in seven pairs.
The forfeit flips the matrix. It did not raise Keep. The Trust gap is mixed with order.
Verification: commit c92006c; Colab Runtime, then Run all; data/session-01.csv to session-07.csv.
Equilibria are computed. The 84 rounds are observed and descriptive, not a population effect.

03 Impact & dialogue

Practical impact: a real-world case

Case: a screen that copies the buyer's priority and tone.
Show the card first. Do not let SAME waive the forfeit.
The buyer gains only if Keep rises. The Promiser pays 5 after Break.
The matrix makes Break worse. Seven pairs did not raise Keep.
The flip fails if Break pays more than 11, or collection fails.

Interdisciplinary evidence

Behavioral science · Hugging Face
Seven pairs, 84 rounds. Trust 26/28 forfeit, 18/28 without.
Descriptive only: forfeit cards are rounds 5, 7, 11, and 12.
Keep near half. SAME 28/42; DIFFERENT 32/42 (DeBruine 2002).
Always Verify scores 24. These Deciders averaged 19.7.
Computer science · GitHub
Nashpy matches. Twelve tests pass. Seed 206: 48 / 24 / 22 / 18.
Commit c92006c. Colab Run all. Clicks are not these sessions.

Limitations & next test

Keep stayed near one half, and the Trust gap is mixed with round order.
Next test: shuffle the card order, and record consent before treating scores as behavior.
If Keep stays near one half, should a buyer still post the forfeit?

SEP 28 Symposium comments were on a withdrawn draft

SEP 30 Final revision shown on this poster

Acknowledgments

I thank Professor Luyao Zhang. The PS1 review asked for two acting players and visible matrices. Sole author: Muhan Chen. Cursor, 29-30 Sep 2026, implemented a model I had locked. I did not invent the seven pairs.

Selected references

Nash, J. F. (1950). Equilibrium points in n-person games. PNAS, 36(1), 48-49. <https://doi.org/10.1073/pnas.36.1.48>
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Crawford, V. P., and Sobel, J. (1982). Strategic information transmission. Econometrica, 50(6), 1431-1451. <https://doi.org/10.2307/1913390>
DeBruine, L. M. (2002). Facial resemblance enhances trust. Proc. R. Soc. B, 269, 1307-1312. <https://doi.org/10.1098/rspb.2002.2034>

OPEN MATERIALS

Hugging Face
https://huggingface.co/spaces/dku-comsci-econ206-2026/Trust_Forget
GitHub · commit c92006c
<https://github.com/dku-comsci-econ206-Autumn2026/PS2-Muhan>

